

PIR Security Light

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Presented here is a simple passive infrared (PIR) sensor security light that turns on automatically when someone comes close to it. Dark places in your home or office make perfect hiding places for burglars and thieves, and in such situations this security light is very useful.

Fig. 1 shows the author's prototype of the PIR security light.

A PIR sensor is an electronic device that detects the infrared light radiated from objects in its field of view. All objects with temperature above zero absolute (0°K) emit what is known as 'black-body radiation.' This energy is invisible to the human eye but can be detected by electronic devices designed



Fig. 1: Author's prototype

Test Points

Test point	Details
TP0	0V, GND
TP1	9V
TP2	8.3V approx.
TP3	3.3V, when PIR is activated
TP4	9V, when PIR is activated

PARTS LIST

Semiconductors:	
T1	- BC547 npn transistor
T2	- BD140 pnp transistor
LED1-LED16	- 5mm white LED
D1	- 1N4001 rectifier diode
CON1	- PIR sensor
Resistors (all ¼-watt, ±5% carbon):	
R1, R9-R10	- 1-kilo-ohm
R2	- 10-kilo-ohm
Miscellaneous:	
BATT.1	- 9V PP3/6F22 battery
S1	- On/off switch

for such a purpose. When an infrared source with above-zero temperature, such as a human being, comes near the PIR sensor, the apparent motion is instantly detected.

Circuit and working

Fig. 2 shows the circuit of PIR security light. The circuit is built around a PIR sensor and a few discrete components. The PIR sensor is a pyroelectric sensor module developed for the detection of human body radiations. The PIR sensor module has an instrumentation circuit on board that amplifies the signal to appropriate voltage level so as to indicate the detection of motion.

The PIR sensor, combined with a Fresnel lens, is mounted on a compact-size PCB along with an analogue IC and some passive components to form the module (shown in Fig. 5). With this, a high-level output of variable width is provided. The PIR sensor has a single output that goes high when a valid motion is detected.

Usually, PIR sensor modules have a 3-pin connection: V_{cc} , output and GND. The pin-out may vary, so it is recommended to check the manufacturer's datasheet in order to confirm the pins. Sometimes, they

have markings on the board next to the pins. Besides, PIR sensor modules also have a 2-pin jumper selection for single or continuous trigger output mode. The two positions are labelled as 'H' and 'L.' When the jumper is at position 'H,' the output remains high whenever the sensor is re-triggered repeatedly. In position 'L,' the output goes high and low every time the sensor is triggered. So a continuous motion will give repeated high/low pulses in this mode.

The circuit presented here continuously monitors the output from the PIR sensor module and turns a bunch of white LEDs on when the sensor output goes high. When switch S1 is turned on, 9V DC supply from the 9V battery (BATT.1) enters the whole circuit. The circuit automatically switches to 'active' mode for a while and then shifts back to 'sleep' mode. During 'sleep' mode, power consumption is very low.

The output of the PIR sensor module is connected to the base of transistor T1 (BC547) through resistor R1. When any motion is detected, PIR sensor output goes high to about 3.3V. (The sensor module has a 3.3V regulator IC on board.) During the normal condition ('sleep' mode), transistor T1 is cut-off, and the collector output is at a high level (+9V). When motion is sensed, the high output from the sensor module saturates transistor T1 and the voltage at its collector drops down to activate transistor

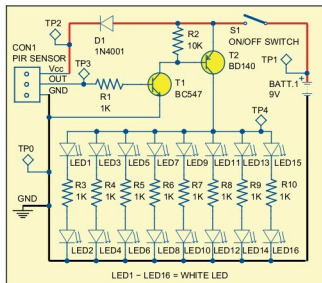


Fig. 2: Circuit of PIR security light

